Predicted structures and the numbers that say do not trust me

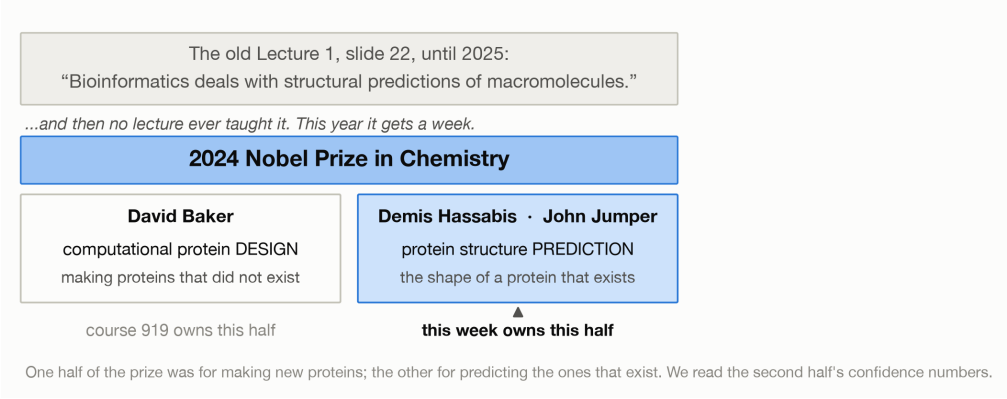
- Öngörülen yapılar ve "bana güvenme" diyen sayılar
- BIB Week 12

Ad of the Week #12

- Skills Ledger: tick structure prediction and model quality assessment

A promise this course used to break

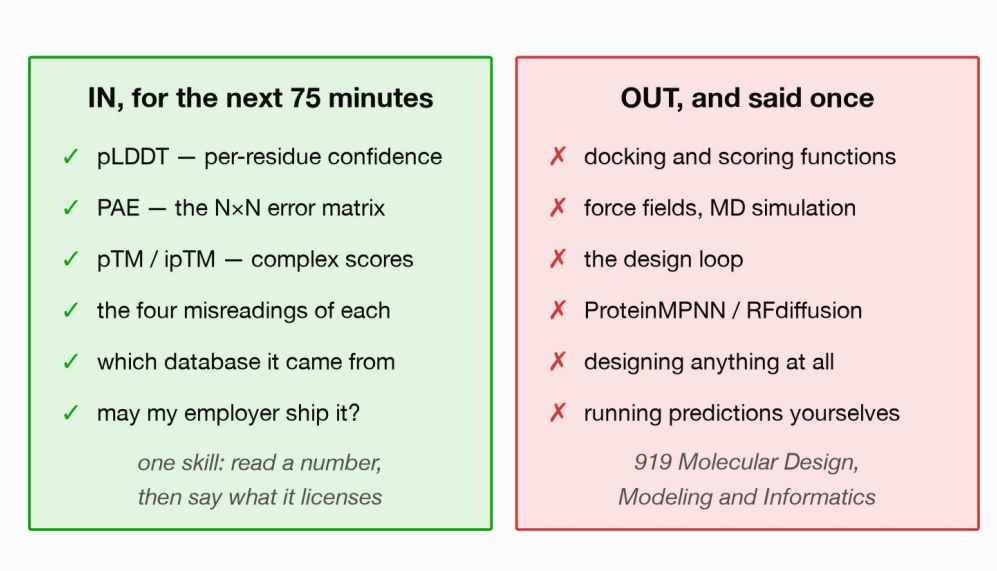
- Until 2025, Lecture 1 said: "Bioinformatics deals with structural predictions of macromolecules"
- And then no lecture ever taught it. This year it gets a week.
- 2024 Nobel Prize in Chemistry: Baker, for computational protein design
- …and Hassabis and Jumper, for protein structure prediction (yapı öngörüsü)
- A vague promise becomes a dated fact — and the dated fact is now two years old
- What you get today is not the model. It is the four numbers it prints beside the model.



The old Lecture 1 promise, and the 2024 Nobel Prize in Chemistry that dates it

What this week is, and what it is not

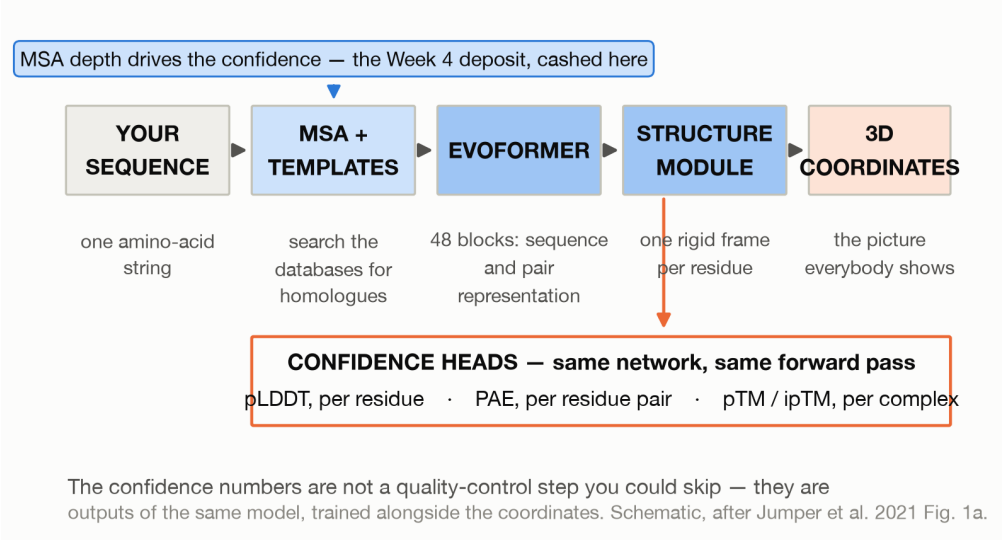
- In: pLDDT, PAE, pTM and ipTM — four numbers, and the four ways each is misread
- In: which database an object came from, and whether your employer may legally ship the model
- Out: docking, force fields, molecular dynamics, the design loop, ProteinMPNN, RFdiffusion
- Those belong to 919 Molecular Design, Modeling and Informatics, in this same programme
- You will not run a prediction this week. There is no AlphaFold Server job with your name on it.
- One skill, examinable: read a confidence number and say what it does and does not license



In: reading four numbers. Out: everything course 919 owns.

AlphaFold 2 in one diagram

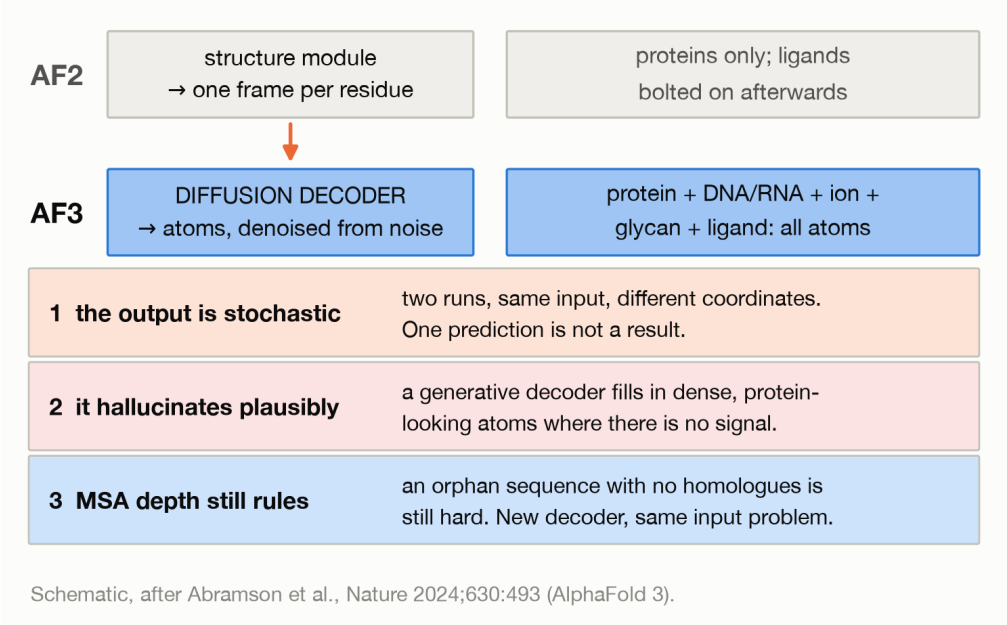
- Input is a sequence. The first thing the model does is go looking for relatives.
- MSA (çoklu dizi hizalaması) and templates → Evoformer (48 blocks) → structure module → coordinates
- The structure module emits one rigid frame per residue — position and orientation
- The confidence heads are part of the same network, trained alongside the coordinates
- So pLDDT and PAE are not an external quality check. They are predictions too.
- Week 4 deposit, cashed here: MSA depth drives the confidence



AF2: sequence → MSA → Evoformer → structure module → coordinates, with the confidence heads hanging off the same network

AlphaFold 3 changed the decoder, not the problem

- AF3 replaces the structure module with an atom-level diffusion decoder (difüzyon çözücü)
- Everything is now atoms: protein, DNA, RNA, ions, glycans, ligands — one representation
- 1. The output is stochastic (olasılıksal): two runs, same input, different coordinates
- 2. A generative decoder hallucinates plausibly — dense, protein-looking atoms where there is no signal
- 3. MSA depth still governs orphan sequences. New decoder, same input problem.
- Therefore: one prediction is not a result. Write that down; complexes will make it bite.



AF2's structure module becomes AF3's diffusion decoder — and the three consequences

CASP16: one task is finished, four are not

- Single-domain fold prediction is essentially solved. That is a real and remarkable result.
- Not solved: complexes and interfaces · stoichiometry · truncated constructs
- Not solved: conformational change induced by binding a partner
- Notice that all four open problems are the ones a wet lab actually asks about
- "AlphaFold solved protein folding" is a headline. The assessors' verdict is narrower and more useful.

One task is finished. Four are not — and all four are the ones you will be asked for.

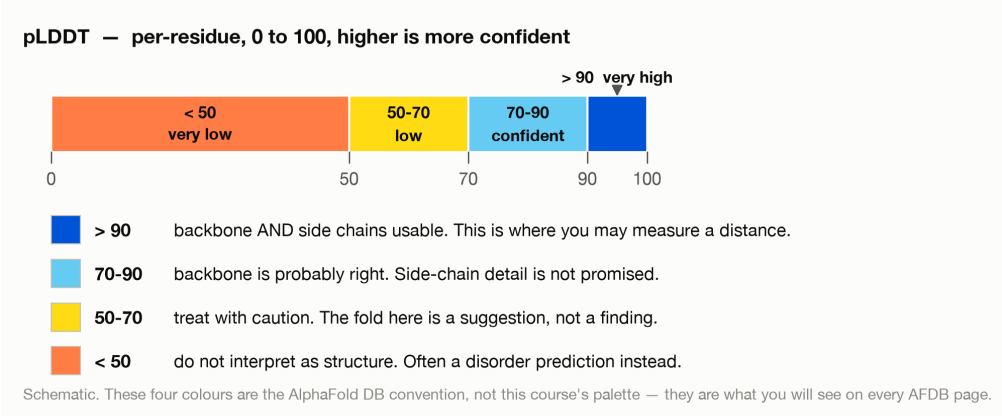
the task	CASP16 verdict	what you therefore do
single-domain fold prediction	**essentially solved**	trust the fold; still read pLDDT
complexes and interfaces	**NOT solved**	run seeds; read ipTM; add a decoy
stoichiometry (copy number)	**NOT solved**	you supplied it — the model agreed
truncated constructs	**NOT solved**	it folds what you gave it, confidently
interaction-induced change	**NOT solved**	one static answer to a dynamic question

Summary of the CASP16 assessment as reported by the organisers; wording is the instructor's.

CASP16: what is finished, what is not, and what you therefore do

pLDDT: one number per residue

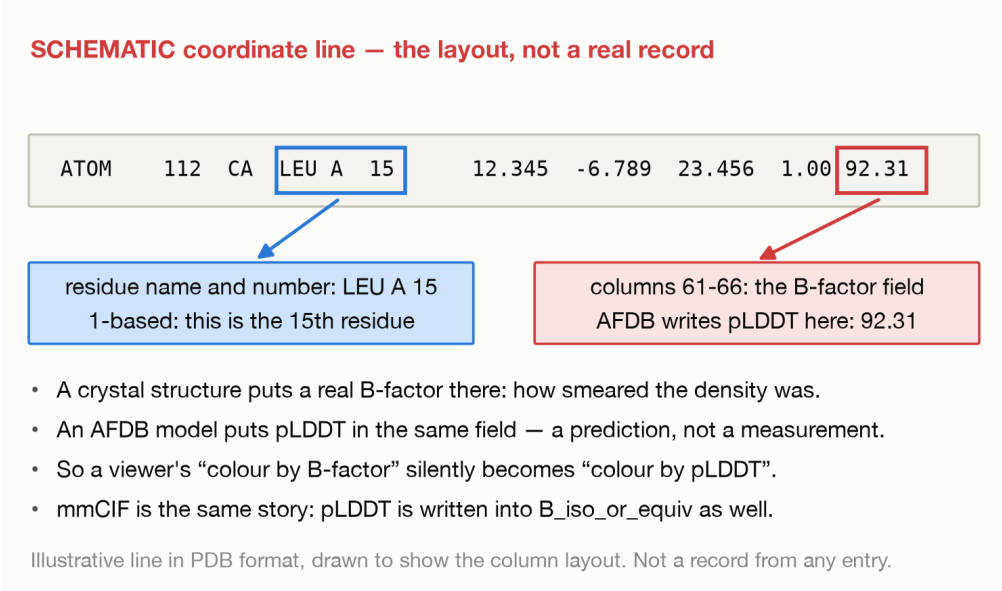
- pLDDT = predicted lDDT-Cα: the model's own estimate of local accuracy, 0 to 100, per residue
- Four bands, and they are the AFDB convention you will meet everywhere:
  - > 90 very high · 70–90 confident · 50–70 low · < 50 very low
- Dark blue / light blue / yellow / orange — memorise the colours, not just the numbers
- It is local: a residue's pLDDT says nothing about where that residue sits relative to a distant one
- An average pLDDT over a whole protein can hide a perfectly folded domain and a 40-residue mess



The four AFDB pLDDT bands, with what each one licenses

Where the number physically lives

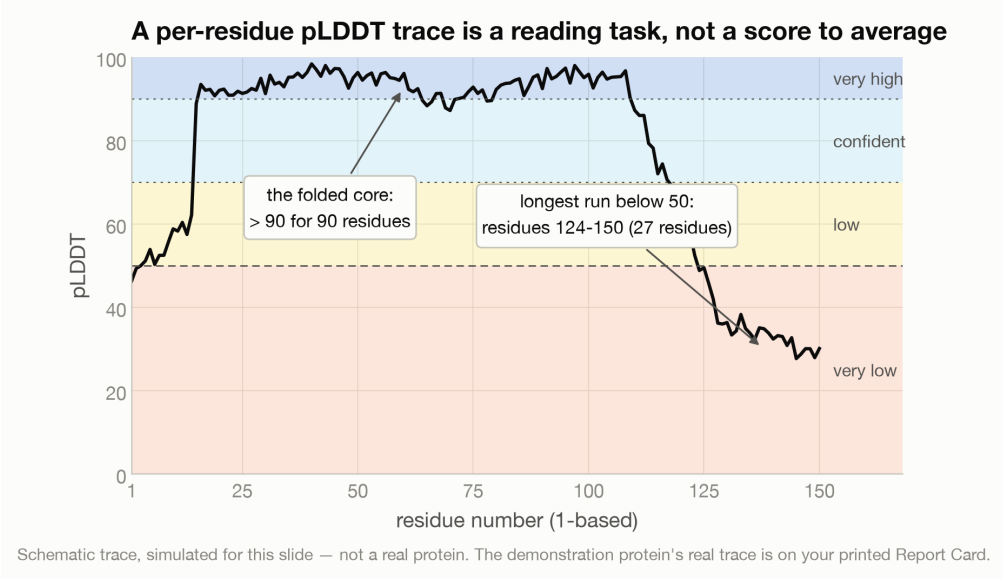
- AFDB writes pLDDT into the B-factor column of the coordinate file (columns 61–66 in PDB format)
- A crystal structure puts a real B-factor there: how smeared the electron density was
- Same column, different object. One is a measurement; the other is a prediction.
- So "colour by B-factor" in any viewer silently becomes "colour by pLDDT" — and nothing warns you
- In mmCIF it is written into B_iso_or_equiv as well, so the trap survives the format change
- GENOMES S row: name the file, the database and the date you downloaded it



Where pLDDT lives: the B-factor column of a schematic coordinate line

Read the trace, not the average

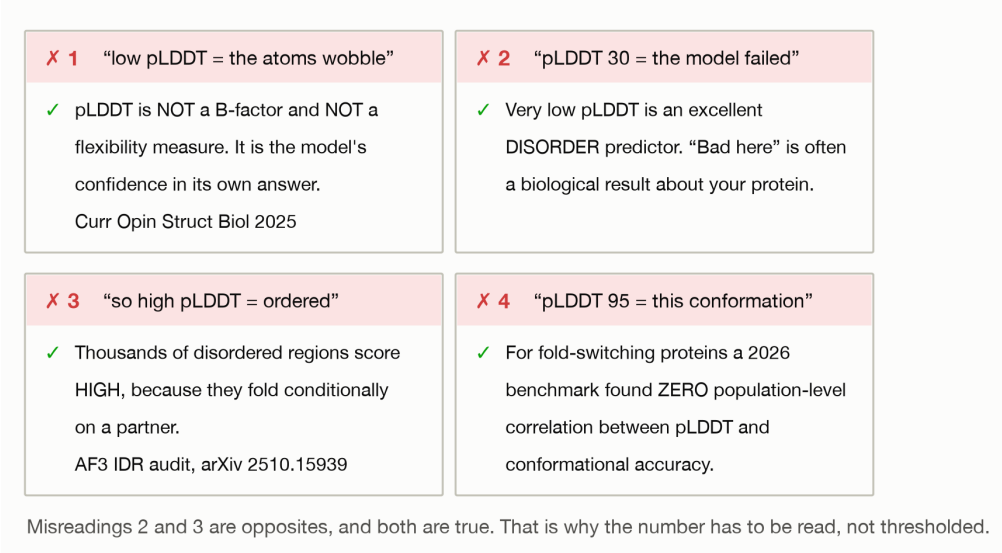
- The object you actually read is a per-residue trace: pLDDT against residue number
- A typical eukaryotic protein: a confident core and a disordered tail (düzensiz kuyruk)
- Find the longest run below 50 and write down its first and last residue, as integers
- Then ask the only question that matters: is this region a model failure, or a biological result?
- You cannot answer that from the trace alone. You answer it with an annotation (DisProt, UniProt).
- This afternoon you do exactly this, on paper, on one printed trace



Schematic pLDDT trace: a confident core, a linker, and a long disordered tail

Four ways to misread pLDDT

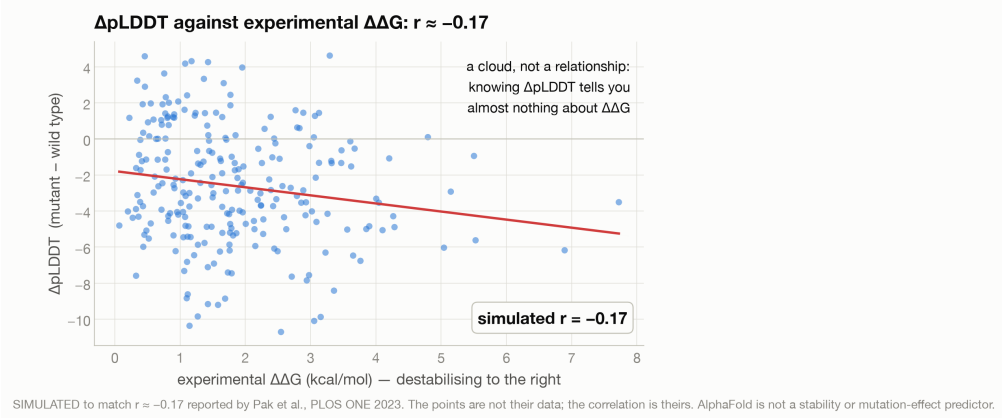
- 1. It is not a B-factor and not a flexibility measure (Curr Opin Struct Biol 2025)
- 2. Very low pLDDT is an excellent disorder predictor — "bad here" is often a biological result
- 3. But high pLDDT does not mean ordered: many IDRs score high because they fold on a partner
- 4. High pLDDT does not pick a conformation: for fold-switching proteins, no correlation at all
- Misreadings 2 and 3 are opposites and both are true. That is why it is read, not thresholded.
- Examinable: given a described region and its pLDDT, name which misreading is in play



The four misreadings, as a family of four

The negative result that Week 13 needs

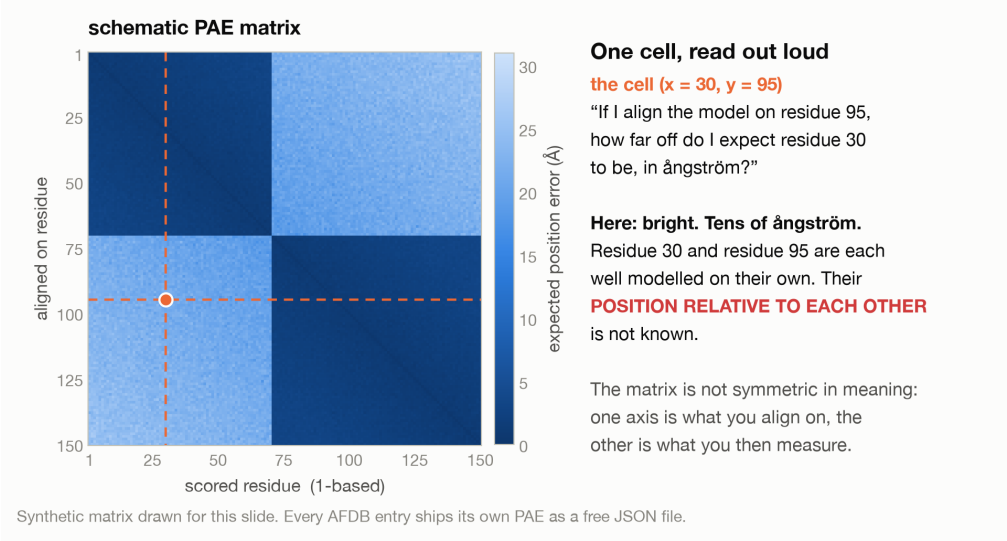
- Wild type and mutant predicted separately, ΔpLDDT compared against experimental ΔΔG
- r ≈ −0.17 — Pak MA et al., PLOS ONE 2023
- In words: knowing how much the confidence dropped tells you almost nothing about stability
- AlphaFold is not a stability predictor and not a mutation-effect predictor. Bank this sentence.
- It was never trained to be one. A point mutation barely perturbs the MSA it reads.
- Week 13 cashes this when you meet calibrated variant predictors and the one-tool rule



Simulated to match r ≈ −0.17: ΔpLDDT against experimental ΔΔG

PAE: the matrix, in ångström

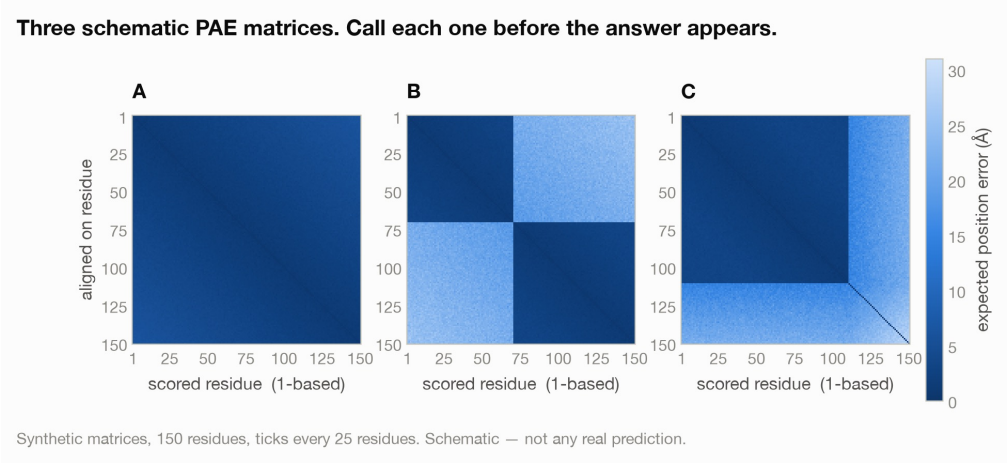
- PAE = predicted aligned error: an N×N matrix (matris), one value per ordered pair of residues
- Cell (x, y) = "if I align the model on residue y, how far off do I expect residue x to be?"
- The unit is ångström, so the number means something physical — not a score from 0 to 1
- Dark = low expected error = confident. Pale = tens of ångström = unknown.
- The diagonal is always dark and tells you nothing. The information is off-diagonal.
- Every AFDB entry ships its PAE as a free JSON file. You never have to run anything to get it.



One cell of a schematic PAE matrix, read out loud

Three matrices. Call each one before I speak.

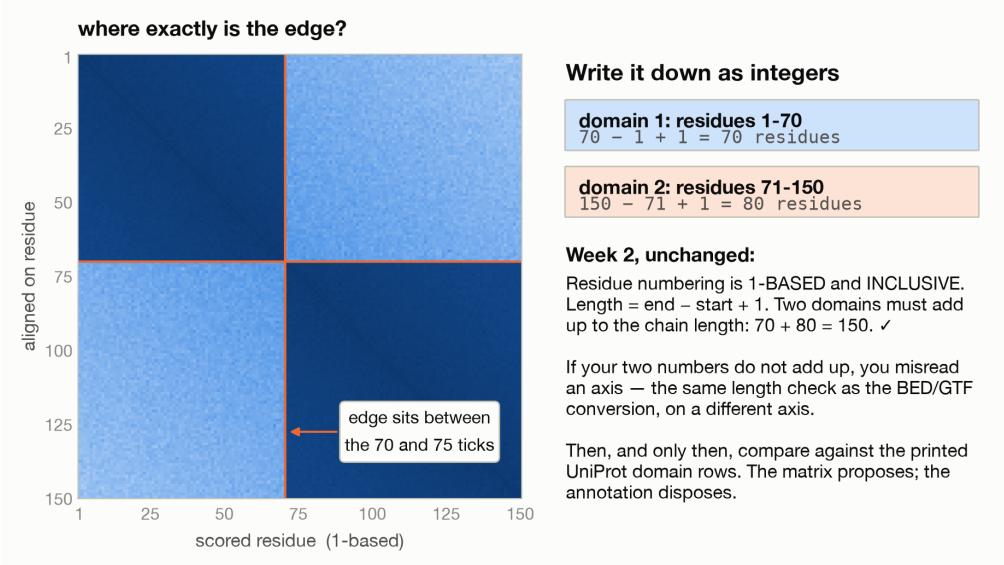
- A — dark everywhere: one compact domain, the whole thing placed consistently
- B — two dark blocks, bright off-diagonal: two confident domains, relative placement unknown
- C — a bright cross: one confident core plus a disordered tail that goes nowhere definite
- The drill rule: high pLDDT everywhere PLUS a bright off-diagonal block…
- …= two well-modelled domains whose relative arrangement is unknown
- That rule is examinable, and it is the single most useful thing in this lecture



Three schematic PAE matrices for the room to call: compact, two domains, disordered tail

From a block edge to a residue number

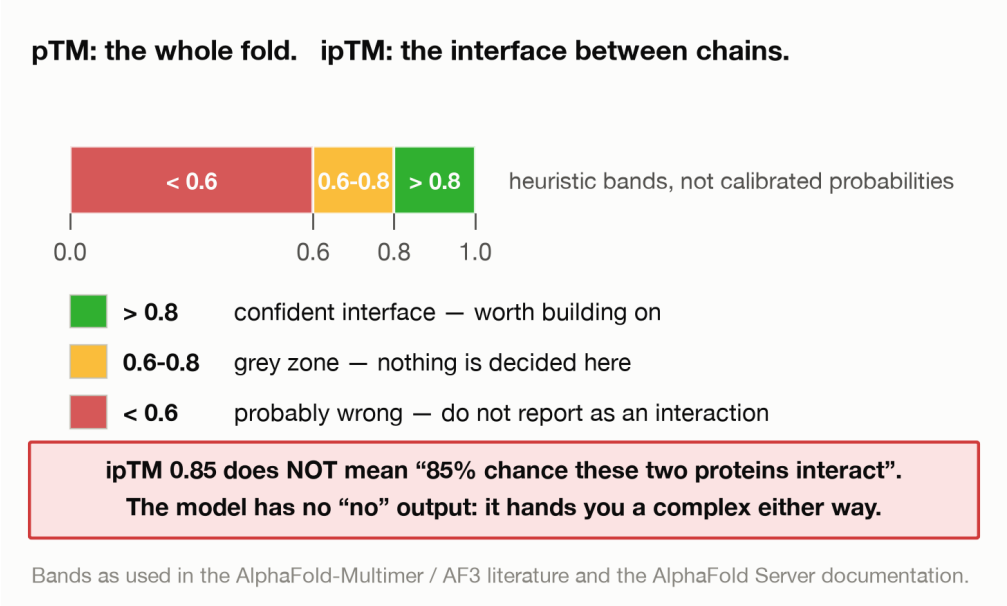
- The question is not "where is the boundary?" It is "what integer do I write down?"
- Ticks every 25 residues on both axes, so converting an edge to a number is reading, not indexing
- Residue numbering is 1-based and inclusive — Week 2, unchanged
- Domain 1 = residues 1–70 → length 70 − 1 + 1 = 70. Domain 2 = 71–150 → 80. Sum = 150. ✓
- The length check is the control: if your two domains do not add to the chain length, you misread an axis
- Then — and only then — compare against the printed UniProt domain rows



Turning a visual block edge into residue integers, with the Week 2 length check

Complexes: pTM and ipTM

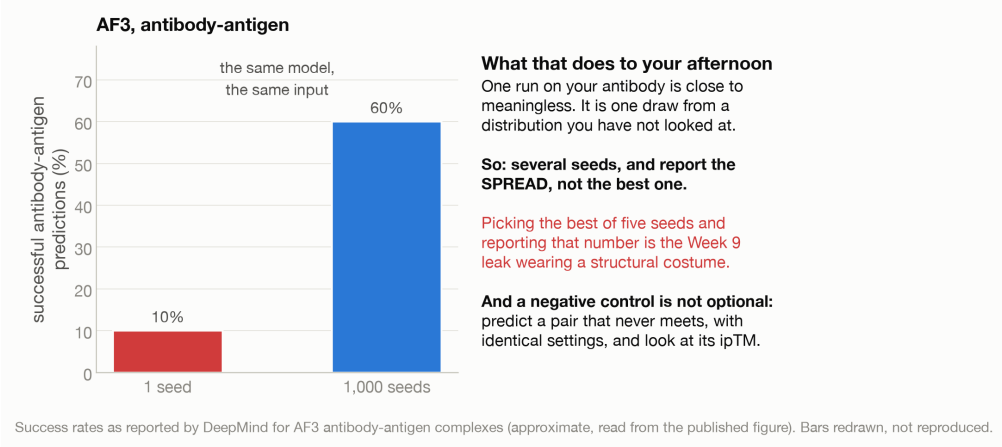
- pTM: one number for the whole predicted fold. ipTM: one number for the interface between chains.
- Heuristic bands: > 0.8 confident · 0.6–0.8 grey zone · < 0.6 probably wrong
- They are rules of thumb from the literature, not calibrated probabilities
- ipTM is not the probability that two proteins interact. It is the most misquoted number here.
- Two proteins that never meet in a cell are still handed a complex: the model has no "no" output
- So a negative control (negatif kontrol) is not optional — it is the only thing that makes ipTM readable



pTM and ipTM bands, and the sentence they do not license

One seed is not a result

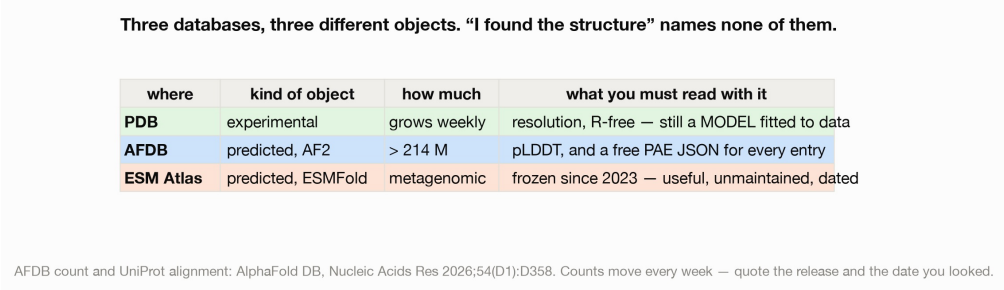
- AF3 antibody–antigen success: about 10% at one seed, about 60% at 1,000 seeds
- DeepMind's own figure. Same model, same input — the difference is how many times you asked.
- So one run on your antibody is close to meaningless — and best-of-five reported as one number is worse
- Report the spread across seeds, not the best one. Best-of-N is the Week 9 leak in a new costume.
- The control that makes a number readable: the same prediction on a pair that never meets
- Six pre-run jobs this afternoon: a true complex, a decoy, and three seed replicates



Antibody–antigen success at 1 seed versus 1,000 seeds, and what it costs you

Three databases, three different objects

- PDB — experimental, yet still a model fitted to data: hence resolution and R-free
- AFDB — predicted, aligned to UniProt, > 214 million entries, each with pLDDT and PAE
- ESM Atlas — ESMFold predictions at metagenomic scale, frozen since 2023
- "I found the structure of my protein" names none of the three — they are not interchangeable
- GENOMES S and E: database, accession, version and the date you looked
- Experimental is not automatically better: check the resolution and what was crystallised



Three databases, three different objects. "I found the structure" names none of them.

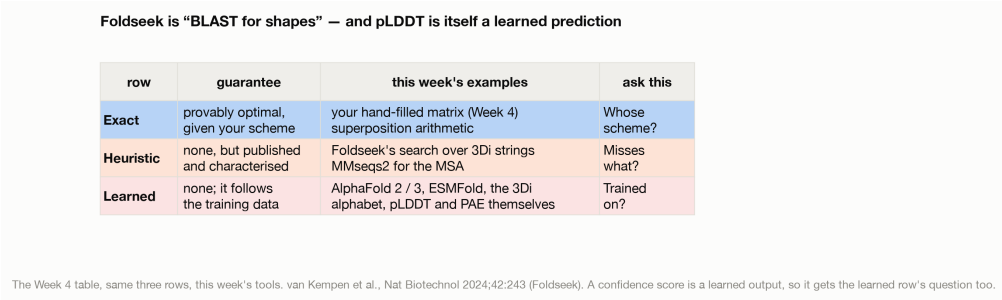
where	kind of object	how much	what you must read with it
PDB	experimental	grows weekly	resolution, R-free — still a MODEL fitted to data
AFDB	predicted, AF2	> 214 M	pLDDT, and a free PAE JSON for every entry
ESM Atlas	predicted, ESMFold	metagenomic	frozen since 2023 — useful, unmaintained, dated

AFDB count and UniProt alignment: AlphaFold DB, Nucleic Acids Res 2026;54(D1):D358. Counts move every week — quote the release and the date you looked.

Three databases, three different objects, and what you must read with each

Foldseek: BLAST for shapes

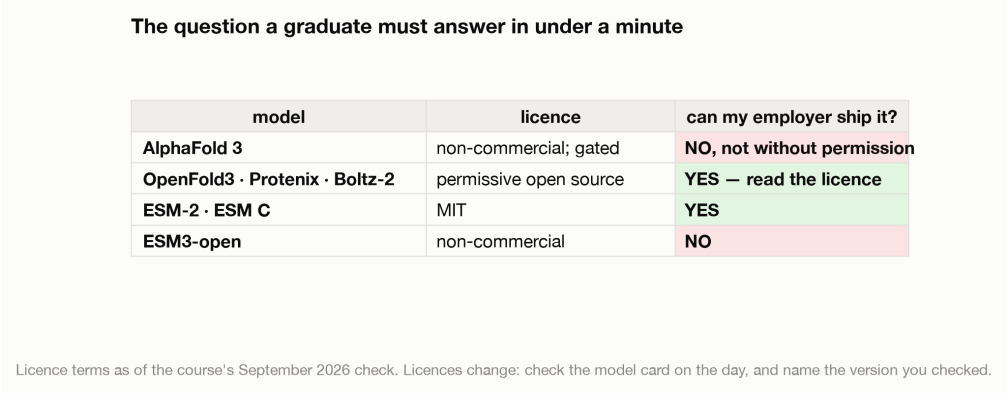
- Week 4's trichotomy, continued: exact / heuristic / learned (kesin / sezgisel / öğrenilmiş)
- Foldseek turns a 3D structure into a string over a learned structural alphabet (3Di), then searches it
- Learned representation, classical search — the same trick, on a different object
- van Kempen et al., Nat Biotechnol 2024;42:243 — and it is fast enough to search all of AFDB
- The converse the old slides never taught: similar structure does not require similar sequence
- pLDDT and PAE are learned outputs too, so they get the learned row's question: trained on what?



The Week 4 trichotomy with this week's tools dropped into the rows

Licences: can my employer ship this?

- AlphaFold 3 — non-commercial, weights gated on request. The answer is NO without permission.
- OpenFold3, Protenix, Boltz-2 — permissive open-source licences. Usually YES; read the exact terms.
- ESM-2 and ESM C — MIT. ESM3-open — non-commercial. Same family, different answers.
- Free to use in a browser says nothing about whether you may build a product on it
- You will be asked this in your first job, with about a minute to answer
- Licence (lisans) terms change: name the model, the version and the date you checked



The question a graduate must answer in under a minute

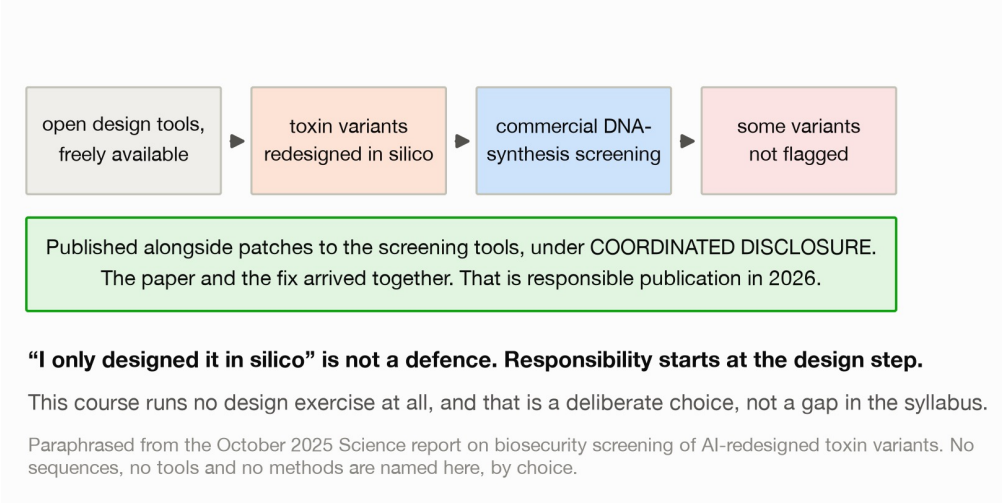
model	licence	can my employer ship it?
AlphaFold 3	non-commercial; gated	**NO, not without permission**
OpenFold3 · Protenix · Boltz-2	permissive open source	**YES — read the licence**
ESM-2 · ESM C	MIT	**YES**
ESM3-open	non-commercial	**NO**

Licence terms as of the course's September 2026 check. Licences change: check the model card on the day, and name the version you checked.

The licence table: model, licence, and whether your employer may ship it

The dual-use beat

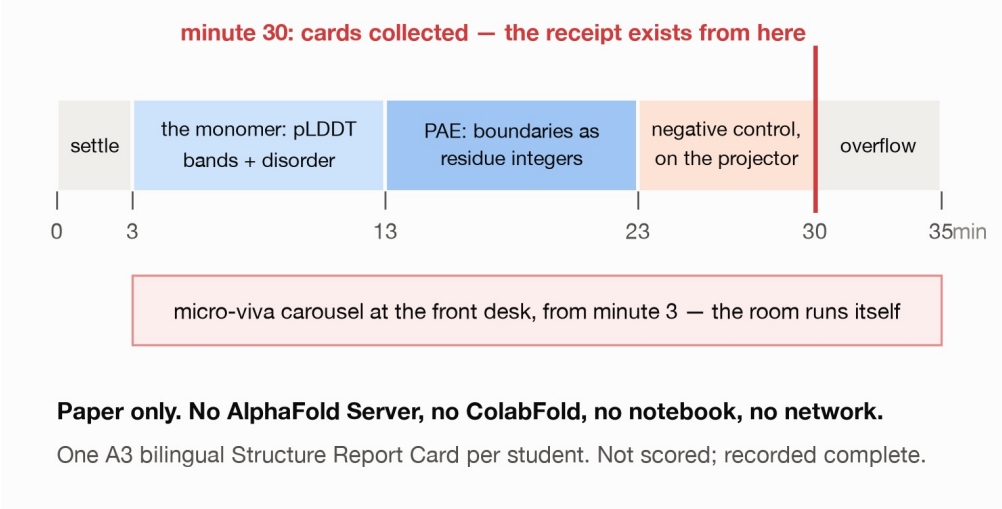
- October 2025, Science: open protein-design tools were used to produce redesigned toxin variants
- Some of those variants were not flagged by commercial DNA-synthesis biosecurity screening
- It was published together with patches to the screening tools, under coordinated disclosure
- That pairing is the model: the fix was built with the screening providers before the hole went public
- "I only designed it in silico" is not a defence. Responsibility starts at the design step.
- This course runs no design exercise at all — a deliberate choice, not a gap in the syllabus



The paraphrase study, and the coordinated disclosure that came with it

This afternoon: the Structure Report Card

- 35 minutes, paper only. No AlphaFold Server, no ColabFold, no notebook, no network.
- One printed bilingual A3 Structure Report Card each. The micro-viva carousel starts at minute 3.
- Minutes 3–13: mark the bands on the printed trace, find the longest run below 50, check the disorder row
- Minutes 13–23: read the PAE, write the domain boundaries as residue integers, check the lengths add up
- Minutes 23–30: the negative control, worked once on the projector — read ipTM, compute the seed spread
- Cards collected at minute 30. Same ritual and same slot as an A.finish(). Not scored.



The 35-minute paper hour: carousel from minute 3, cards collected at minute 30